Supplementary Information

HBALS: A Hybrid Bees Algorithm with 2-opt Local Search for the Travelling Salesman Problem, Evaluated by Ablation and Equal-Budget Comparison

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Supplementary Table S1. Settings of the compared methods

Method	Description and settings
HBALS	Final configuration (main text, Table 4).
ILS-2opt	Nearest-neighbour start (random city), double-bridge kick, 2-opt to a local optimum (no move limit), acceptance if not worse.
GA-2opt	Population 20, tournament size 3, order crossover (0.8), swap mutation (0.1), elitism 2, 2-opt (at most 20 moves) on every offspring.
ACO-2opt	Ant system: 20 ants, $\alpha = 1$, $\beta = 2$, evaporation 0.5, $Q = 100$, initial pheromone 0.1, 2-opt (at most 20 moves) on every ant tour.
MS-2opt	Multi-start: random tour, 2-opt to a local optimum, keep the best.
BA	Plain bees algorithm: random initialisation, m = 12, e = 3, nep = 8, nsp = 4, three single-swap neighbours per bee, scouts, no local search; swap moves evaluated in constant time.

All 2-opt calls use the same first-improvement operator with restart after each improving move.

Supplementary Table S2. Instances and evaluation budgets

Instance	N	Best-known length	Budget (million move evaluations)
eil51	51	426	38.5
berlin52	52	7542	42.1
st70	70	675	85.1
eil76	76	538	84.3
kroA100	100	21282	174.5
kroB100	100	22141	166.6
eil101	101	629	186.1
lin105	105	14379	228.3
ch130	130	6110	338.2
ch150	150	6528	429.6
rat195	195	2323	755.6
d198	198	15780	754.4
kroA200 (larger)	200	29368	772.3
a280 (larger)	280	2579	1790.6
lin318 (larger)	318	42029	1994.4

Supplementary Table S3. HBALS versus ILS-2opt on each of the 12 main instances

Instance	HBALS mean gap (%)	ILS mean gap (%)	Runs at best-known length (HBALS / ILS)	A12	Raw p	Holm-adjusted p
eil51	0.035	0.200	17 / 7	0.77	0.001	0.011
berlin52	0.000	0.154	20 / 19	0.53	0.342	1.000
st70	0.030	0.348	17 / 14	0.60	0.154	1.000
eil76	0.158	0.316	12 / 14	0.48	0.836	1.000
kroA100	0.008	0.117	18 / 7	0.79	0.0003	0.004
kroB100	0.237	0.351	4 / 3	0.67	0.064	0.514
eil101	0.175	0.350	8 / 5	0.55	0.553	1.000

Instance	HBALS mean gap (%)	ILS mean gap (%)	Runs at best-known length (HBALS / ILS)	A12	Raw p	Holm-adjusted p
lin105	0.160	0.208	10 / 10	0.55	0.613	1.000
ch130	0.601	0.676	1 / 1	0.51	0.903	1.000
ch150	0.545	0.541	0 / 2	0.43	0.432	1.000
rat195	1.184	0.837	0 / 0	0.24	0.006	0.055
d198	0.461	0.324	0 / 0	0.27	0.015	0.134

Two-sided Mann–Whitney U tests on n = 20 runs per method; A12 is the probability that a run of HBALS is shorter than a run of ILS-2opt (0.5 means no difference); Holm adjustment over the 12 instances; $\alpha = 0.05$.

Supplementary Table S4. Paired comparisons of HBALS with each method over the 12 instances

Comparator	HBALS better on	Comparator better on	Raw p	Holm-adjusted p
ILS-2opt	9	3	0.129	0.129
GA-2opt	12	0	0.0005	0.002
ACO-2opt	12	0	0.0005	0.002
MS-2opt	12	0	0.0005	0.002
BA	12	0	0.0005	0.002

Two-sided Wilcoxon signed-rank tests on the 12 per-instance mean gaps (n = 12 pairs); Holm adjustment over the five comparisons; $\alpha = 0.05$. The Friedman test over the six methods gave $\chi^2(5) = 55.0$, $p = 1.3\times10^{-10}$; the Nemenyi critical difference is 2.18.

Supplementary Table S5. HBALS with early stopping and with the full budget

Instance	Mean iterations (early stop)	Mean gap, early stop (%)	Mean gap, full budget (%)	Mean iterations (full budget)
eil51	38	0.481	0.035	306
berlin52	41	0.010	0.000	299
st70	52	0.526	0.030	325
eil76	52	1.208	0.158	281
kroA100	72	0.546	0.008	291
kroB100	65	0.603	0.237	305
eil101	58	1.312	0.175	304
lin105	74	0.983	0.160	311
ch130	94	1.223	0.601	320
ch150	79	1.322	0.545	301
rat195	73	2.191	1.184	295
d198	92	1.010	0.461	295

Early stopping: stop after 20 consecutive iterations without improvement of the best tour, at most 300 iterations; n = 20 runs per cell.

Supplementary Table S6. HBALS and values reported in the literature for bees-algorithm variants

Instance	Method	Best	Mean	Mean gap (%)
eil51 (best-known 426)	HBALS (this study)	426	426.2	0.04
	Suluova & Pham (2024)	437	437	2.58
berlin52 (best-known 7542)	HBALS (this study)	7542	7542.0	0.00
	Sahin (2023)	7544	7544	0.03
	Suluova & Pham (2024)	7930	7930	5.14

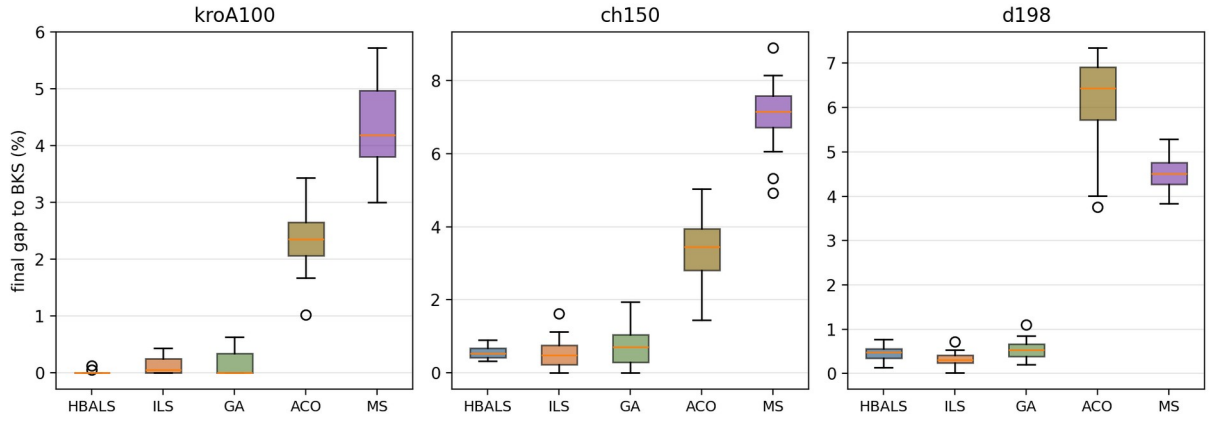
Instance	Method	Best	Mean	Mean gap (%)
st70 (best-known 675)	HBALS (this study)	675	675.2	0.03
	Sahin (2023)	677	677	0.31
	Suluova & Pham (2024)	696	696	3.11
kroA100 (best-known 21282)	HBALS (this study)	21282	21283.8	0.01
	Sahin (2023)	21285	21285	0.02
	Suluova & Pham (2024)	22018	22018	3.46
ch150 (best-known 6528)	HBALS (this study)	6549	6563.5	0.54
	Sahin (2023)	6552	6554	0.37
	Suluova & Pham (2024)	6928	6928	6.13
rat195 (best-known 2323)	HBALS (this study)	2338	2350.5	1.18
	Sahin (2023)	2340	2352	0.73

Literature values were taken from the original publications and were not re-run; budgets, hardware and stopping rules differ, so the comparison is indicative only. Sahin did not report eil51, and Suluova and Pham did not report rat195.

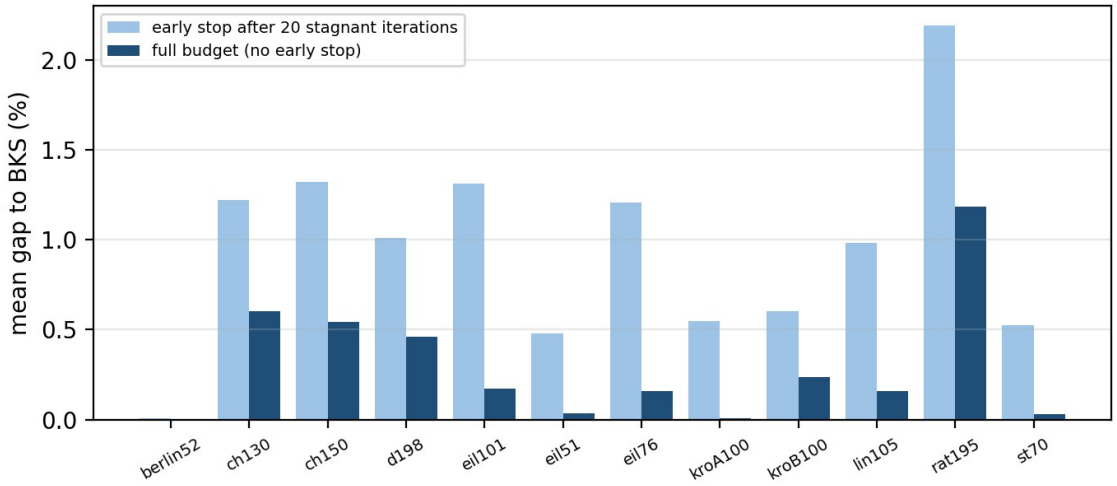
Supplementary Table S7. Absolute tour lengths of HBALS and ILS-2opt

Instance	Best-known	HBALS best	HBALS mean	HBALS s.d.	ILS best	ILS mean	ILS s.d.
eil51	426	426	426.2	0.4	426	426.9	0.7
berlin52	7542	7542	7542.0	0.0	7542	7553.6	52.1
st70	675	675	675.2	0.5	675	677.3	4.2
eil76	538	538	538.8	1.3	538	539.7	3.4
kroA100	21282	21282	21283.8	6.1	21282	21306.8	28.2
kroB100	22141	22141	22193.5	38.5	22141	22218.8	43.7
eil101	629	629	630.1	1.1	629	631.2	3.7
lin105	14379	14379	14401.9	32.2	14379	14408.9	35.8
ch130	6110	6110	6146.7	21.9	6110	6151.3	28.3
ch150	6528	6549	6563.5	11.4	6528	6563.3	28.8
rat195	2323	2338	2350.5	5.3	2328	2342.5	10.3
d198	15780	15801	15852.8	27.8	15783	15831.2	25.8

n = 20 runs per method; s.d., standard deviation of the tour length over runs.



Supplementary Figure S1 Distribution of the final gap to the best-known tour length over 20 runs on kroA100, ch150 and d198 (boxes: interquartile range and median; whiskers: 1.5 × interquartile range; plain BA omitted for scale).



Supplementary Figure S2 Mean gap to the best-known tour length for HBALS under early stopping (20 stagnant iterations, at most 300 iterations) and under the full budget; n = 20 runs per bar.